
**Tick By Tick Trading Market Data Interface (API) –
MCX Enhanced Order Book Interface (EOBI) – Version 1.5**

In terms of provisions of the Rules, Bye-Laws and Business Rules of the Exchange and in continuation to Exchange circular no. MCX/CTCL/223/2025 dated April 30, 2025 and MCX/CTCL/057/2024 dated January 29, 2024, Members of the Exchange are notified as under:

Trading Members and Empanelled vendors are requested to note that, the Exchange has released Tick By Tick Trading Market Data Interface API - MCX Enhanced Order Book Interface (EOBI) Version 1.5. The details of changes are mentioned in revision history i.e. Details of template "Trade Report (13201)", please note that there are no changes in EOBI API.

In case of any queries or clarification on new interfaces document, trading members/vendors are requested to get in touch on following contact details:

- Email - apisupport@mcxindia.com
- Phone: +91 22 – 6649 4040 / 6731 8888

Trading Members and Empanelled vendors are requested to take note of the same.

For and on behalf of
Multi Commodity Exchange of India Ltd.

Denny Paul
AVP - Technology
Encl.: As above

Kindly contact Customer Service Team on 022 – 6649 4040 or send an email at customersupport@mcxindia.com for any clarification.



Multi Commodity Exchange of India Limited

Tick By Tick Trading Market Data Interface

-

MCX Enhanced Order Book Interface (EOBI)

Version 1.5
Sept 08, 2025

Copyright

All trademarks that appear in the document have been used for identification purposes only and belong to their respective companies.

Document details

Name	Version no.	Description
MCX_EOBI_API	V 1.5	API documentation for Tick By Tick Interface

Document Revision List

Revision No.	Revision Date	Revision Description
1	22-Jun-2021	Created of base document
2	05-Jul-2023	Added Instrument Info Message in Snapshot block diagram Instrument Summary (13601) : 1 TradeCondition field is added 2. Life time low and Life time high enums are added in >MDEntryType field
3	31-October-2023	Added description for Overview of Supported Message Types and Reserve fields. Modification in Offset value for Index info message-13604
4	02-November-2023	Updated Length of MDEntryType of Instrument Summary Message(13601).
5	25-January-2024	Update Description on Instrument Info(13603) and added new section for Index Info (13604) Update MsgSeqNum field description Index info (13604) Remove implied message Constant from Index info (13604)
6	08-September-2025	Adding template Trade Report (13201)

MCX Contact Details

The Exchange's Member & Vendor may contact Technology Division to seek clarification at:	
Multi Commodity Exchange of India Limited Exchange Square, Suren Road, Chakala, Andheri (East), Mumbai 400 093. www.mcxindia.com	Tel: +91 – 22 – 66494000 / 67318888 Fax: +91 – 22 – 66494151 Email – apisupport@mcxindia.com

Restriction on Use and Disclaimer of Information and Data

All The information contained in this document constitutes a trade secret and/or information that are commercial or financial and confidential or privileged. It is furnished in confidence with the understanding that it will not, without the prior written permission of MCX, be used or disclosed for other than allowed purposes.

The copyright in this work may be vested with MCX and / or its suppliers. No part of this document may be copied, reproduced, stored in a retrieval system, or transmitted, in any form or by any means whether, electronic, mechanical, or otherwise without the prior written permission of MCX.

The recipient acknowledges that MCX and its suppliers may have copyright in the work. The recipient further agrees that the work is confidential information and contains proprietary MCX information belonging to MCX and / or its suppliers. The recipient manifests, by its receipt of the work, its acknowledgment of MCX and / or its suppliers copyright in the work, its acceptance that the work is confidential information, and its compliance with the terms contained in this notice.

Although MCX has made every effort to provide accurate information at the date of publication, it does not give any representations or warranties as to the accuracy, reliability or completeness of the information in this document. Accordingly, MCX, its subsidiaries and their employees, officers and contractors and its suppliers shall not, to the extent permitted by law, be liable for any direct or indirect loss arising in any way (including by way of negligence) from or in connection with anything provided in or omitted from this document or from any action taken, or inaction, in reliance on this document.

MCX reserves the right to amend details in this document at any time and without notice. Strictly for private circulation only. This document must not be circulated to other users without prior permission of MCX.

Contents

1	LIST OF ABBREVIATIONS	6
2	INTRODUCTION	6
3	SERVICE DESCRIPTION	7
3.1	FUNCTIONAL CHARACTERISTICS	7
3.2	TECHNICAL CHARACTERISTICS	7
3.3	ORDER BOOK MANAGEMENT	8
3.4	BUILDING THE ORDER BOOK	9
3.5	ADDING AN ORDER	10
3.6	MODIFYING AN ORDER	10
3.7	DELETING AN ORDER	11
3.8	ORDER EXECUTIONS	11
3.9	TRADE STATISTICS	11
3.10	PRODUCT AND INSTRUMENT STATES	11
3.11	HEARTBEATS	12
3.12	RECOVERY	12
3.13	AVAILABILITY OF ENHANCED ORDER BOOK SERVICE	13
3.14	MESSAGE FORMATS	14
3.15	DATAGRAM STRUCTURE	14
3.16	INCREMENTAL MESSAGES	16
3.17	SNAPSHOT MESSAGES	17
3.18	DATA TYPES	19
4	FUNCTIONAL SPECIFICATION	20
4.1	OVERVIEW OF SUPPORTED MESSAGE TYPES AND RESERVE FIELDS	20
4.2	PACKET HEADER = 13003	21
4.3	HEARTBEAT = 13001	22
4.4	EXECUTION SUMMARY = 13202	23
4.5	ORDER ADD = 13100	24
4.6	TOP OF BOOK = 13504	25
4.7	ORDER MODIFY = 13101	26
4.8	ORDER MODIFY SAME PRIORITY = 13106	27
4.9	ORDER DELETE = 13102	28
4.10	ORDER MASS DELETE = 13103	30
4.11	PARTIAL ORDER EXECUTION = 13105	30
4.12	FULL ORDER EXECUTION = 13104	32
4.13	PRODUCT STATE CHANGE = 13300	33
4.14	MASS INSTRUMENT STATE CHANGE = 13302	35
4.15	INSTRUMENT STATE CHANGE = 13301	38
4.16	PRODUCT SUMMARY = 13600	39
4.17	INSTRUMENT SUMMARY = 13601	41
4.18	SNAPSHOT ORDER = 13602	43
4.19	INSTRUMENT INFO = 13603	44
4.20	TRADE REPORT = 13201	45
5	INDEX	47
5.1	INDEX INFO = 13604	47

1 List of Abbreviations

Abbreviation	Description
EOBI	Enhanced Order Book Market Data Interface
EMDI	Enhanced Price Level aggregated Market Data Interface
ETI	Enhanced Trading Interface
FAST	FIX Adapted for STreaming (FAST Protocol) (FAST ProtocolSM). FIX Adapted for STreaming is a standard which has been developed by the Data Representation and Transport Subgroup of FPLs Market Data Optimization Working Group. FAST uses proven data redundancy reductions that leverage knowledge about data content and data formats
FIX	Financial Information eXchange. The Financial Information eXchange ("FIX") Protocol is a series of messaging specifications for the electronic communication of trade-related messages.
In-band	Incremental and snapshots are delivered in the same channel
Out-of-band	Incremental and snapshots are delivered on different channels.
PMAP	Presence Map
ToB	Top of Book
Simple instruments	Single leg outright contracts
Complex instruments	Any combination of single leg outright contracts, e.g. LTP Based Spread Contracts
T7	T7 trading system developed by Deutsche Börse Group

2 Introduction

The **MCX T7 Enhanced Order Book Interface** (MCX T7 EOBI) provides the entire visible order book, by publishing information on each individual order along with executions and state information in real-time and in an un-netted manner. The interface is available for a selected group of derivatives market benchmark products.

This interface provides greater transparency and efficiency, together with a high throughput at minimal latency. The MCX T7 EOBI disseminates public market data with the following features:

- A full order depth feed; there is no depth restriction.
- Information is sent in form of fixed-length binary messages.
- Intelligent packing of messages into a datagram by including repetitive entities only once in a message.
- Utilization of the widely adopted FIX standard to decrease integration efforts and on-going support costs.
- Dissemination of incremental messages (following state changes) and all Snap-shot messages follow a publishing in sequence based on:
 - Side (bid first, offer second),
 - Price (best price first),
 - Time (highest time-priority first).

The MCX T7 EOBI is designed for participants that rely on **low-latency** at a high throughput with a **high band-width network**. The interface disseminates all visible orders without any depth restriction, when the order books are open, along

with order executions and state information via incremental messages in un-netted manner. Furthermore, snapshot messages always carry existing visible orders without any depth restriction at the time of sending.

Multicast address and port combinations of MCX T7 EOB I are different from netted market data broadcast channel.

3 Service Description

3.1 Functional Characteristics

The MCX T7 EOB I disseminates:

- The instrument identifier, side, price, and quantity of each visible order .
- Trade price and traded quantity for each executed on-exchange trade.
- Order book information disseminated without any depth limitation.
- The trading status of each product and corresponding instruments.
- Intra-day changes regarding complex instruments.
- Recovery via MCX T7 EOB I snapshots.

In order to send public market data as fast as possible, the MCX T7 EOB I publishes only very specific market information. However, participants can derive certain information themselves based on the messages sent out by the MCX T7 EOB I. The following information is not explicitly provided, however can be derived, if needed (from here onwards the term “order” is used to refer both to orders and quotes):

- Price levels; can be derived from individual orders.
- Aggregation at price levels; can be derived from individual orders.
- Information about synthetic prices; can be derived from visible orders received on the MCX T7 EOB I feed.
- Fully matched incoming visible orders; can be derived from execution messages.
- Trade statistics are not provided via the incremental channel to keep the size of messages as small as possible. They can be derived from the order execution messages sent out on the MCX T7 EOB I incremental channel. But, on the other hand, trade statistics are sent out on the MCX T7 EOB I snapshot channel for recovery purposes.

3.2 Technical Characteristics

The MCX T7 EOB I contains similar technical characteristics as the MCX T7 EMDI, such as “Live - Live” multicast, distribution mode and sequence numbering schemes. Anticipating a high load, the size of messages is kept as small as possible.

The following are highlights of the technical characteristics of the MCX T7 EOB I :

- Low-latency multicast for data dissemination with “Live - Live” concept.
- Fixed length optimized message layouts without any compression.
- Uses push-based publishing model in Out-Of-Band distribution mode.
- Packet sequence numbers are incremented per channel only. Additionally, the MarketSegmentID will be provided in the Packet Header only.

- Little Endian and basic data types are used.
- Message padding for better byte alignment.
- Recovery via MCX T7 EOBI snapshot channel as similar to MCX T7 EMDI.

All messages are designed to be as small as possible and are following FIX 5.0 SP2 semantics. The maximum number of bytes per transmission unit (MTU) is limited to 1372 bytes.

All functional and technical reference data information needed for the MCX T7 EOBI is provided in contract master.

3.3 Order Book Management

The MCX T7 EOBI provides an **explicit** message for each order book update by publishing the instrument identifier, side, price, displayed quantity of each visible order in the entire order book, along with the order execution and state information. The order book information will be published for all products which are enabled on MCX T7 EOBI..

An outline of the **visibility** of orders on the MCX T7 EOBI is shown below:

Order Type	Visible in Order book
Regular Limit Order	yes
Triggered Order – Stop Limit Order	yes
Regular Order – GFD / GTC / GTD	yes
Market Orders	yes
Stop Market Order (un-triggered)	no
Stop Limit Order (un-triggered)	no
Regular Limit Order – IOC	no
All types of Rejected Orders	no

Table 1 - Visibility of orders on the MCX T7 EOBI

For each instrument within a product, snapshot messages can be received via the MCX T7 EOBI snapshot channel to build the initial order book. Once the initial order book is built, the order book must be maintained using the corresponding order book updates received on the MCX T7 EOBI incremental channel. On the MCX T7 EOBI incremental channel, order messages are used by participants to maintain the order book, while explicit state change messages are provided to communicate current product and instrument state. Intra-day complex instrument changes will also be communicated via the MCX T7 EOBI incremental channel.

To assist fine filtering and error discovery on the participant side, the MCX T7 EOBI keeps messages in line using a multi-sequencing paradigm. It uses the following two sequencing methods: **packet sequence number** and **message sequence number**.

Packet Sequencing

Each packet on the MCX T7 EOBI feeds is sequenced using contiguous packet sequence numbers. The packet sequence number is incremented for each packet across products on the same feed.

Message Sequencing

In addition to packet sequencing, each product on the MCX T7 EOBI feeds is sequenced contiguously by using message sequencing. This should allow participants to filter products of interest only. The message sequence number is incremented per product across the different message types.

The following sections describe the order book management with respect to the messages sent over the MCX T7 EOBI.

Message layouts can be identified by the *templateID* field which is the (exchange wide) unique identifier for the message layout, and is included in each Message Header. The *templateID* also determines the fixed size of the message.

Message	Template ID
Order Add	13100
Order Modify	13101
Order Modify Same Priority	13106
Order Delete	13102
Order Mass Delete	13103
Partial Order Execution	13105
Full Order Execution	13104
Execution Summary	13202
Top Of Book	13504
Product State Change	13300
Instrument State Change	13301
Product Summary	13600
Instrument Summary	13601
Snapshot Order	13602
Heartbeat	13001
Instrument Info	13603
Trade Report	13201
Index Info	13604

Table 2 - MCX T7 EOBI messages with assigned template IDs

3.4 Building the Order Book

Messages in the MCX T7 EOBI snapshot channels are grouped by product. In order to build an initial order book, participants subscribe to the MCX T7 EOBI snapshot channel. The content of one **snapshot cycle** for one product is described in subsequent sections. The individual orders in the order book are represented in the snapshot message using the Snapshot Order messages. The snapshot messages contain the field *LastMsgSeqNumProcessed* to enable participant synchronization between the MCX T7 EOBI snapshot channel and the MCX T7 EOBI incremental channel.

While subscribed to the MCX T7 EOBI snapshot channel, participants should keep processing incoming data from the MCX T7 EOBI incremental channel. Any incoming incremental messages with a sequence number higher than the *LastMsgSeqNumProcessed* received in the snapshot messages should be applied to the order book after the full snapshot message is processed.

The following data is provided via the MCX T7 EOBI snapshot channel:

- Product State information,
- Instrument State information,
- Trade Statistics per instrument,
- All visible orders in the order book.

During the Continuous Trading instrument state, all visible orders in the order book will be published on the MCX T7 EOBI incremental channel.

As soon as trading is in the state Continuous, all visible orders in the order book will be immediately published on the MCX T7 EOBI incremental channel.

The sequencing of the data in a snapshot cycle is based on the product identifier, the instrument identifier and on the price level. For the product and instrument identifier, the **sending order sequence** is ascending and the orders are sorted from best to worst prices (buy orders are sorted from highest to lowest, and sell orders from lowest to highest).

The visible orders are sent alternating between buy and sell sides, where orders at the same price level are sorted by order time priority from the oldest to the newest order. The visible order book is disseminated per price level in a zig-zag manner, meaning both the sides (Bid and Offer) at each price level are disseminated before moving on to the next price level. If one side providing more orders on the same price level as the opposite side, all orders of the same price level are reprocessed before switching to the next price level.

Assuming the following arbitrary order book is sorted according to imaginary order priority timestamps and order prices where in the orders with the same order prices are sorted according to imaginary order priority timestamps

Buy	Sell
Order _{B1} 100.05	Order _{S1} 100.50
Order _{B2} 100.05	Order _{S2} 100.55
Order _{B3} 99.95	Order _{S3} 100.55
Order _{B4} 99.90	Order _{S4} 100.55
Order _{B5} 99.00	Order _{S5} 101.00
Order _{B6} 97.00	

Picture 2: Order book in a zig-zag manner

As it can be seen from table above, the orders denoted by B1, B2 and S1 are on the first price level. The orders denoted by B3, S2, S3 and S4 are on the second price level. The orders B4 and S5 are on the third price level. In price level fourth and fifth buy orders exist only.

The resulting sending order sequence in zig-zag fashion is: B1, S1 and B2, B3, S2, S3, S4, B4, S5, B5 and B6.

3.5 Adding an Order

An Order Add message will be sent each time a visible order is added to the order book of the corresponding instrument. The message includes the instrument identifier, side, price and displayed quantity of the order.

Information about an incoming order, that matched fully against one or more orders in the order book, can be derived from the associated execution messages or execution summary only.

The remaining part of an incoming order that matches partially will be reported with an OrderAdd message after all associated executions.

3.6 Modifying an Order

If the time-priority, price and/or displayed quantity of an existing order changes, then an OrderModify or Order Modify Same Priority message will be sent.

A modification might result in the order being assigned a new priority timestamp (for example, in the case of a price modification). If it is the case, then an Order Modify message will be sent.

If there is no priority loss with the modification (which may occur for example when quantity is reduced) then the Order Modify Same Priority message will be sent.

3.7 Deleting an Order

When an order is deleted, the MCX T7 EOBI will publish the instrument identifier, side, price and transaction time, i.e., the fields *SecurityID*, *Side*, *Price* and *TransactTime*,

3.8 Order Executions

In order to ease the processing of matches along with the other order book updates by participants the following information is disseminated for each match corresponding to an incoming order:

- first, an execution summary message will be sent when an incoming order has been matched against orders that were already in the order book
- second, messages that convey the individual executions of visible orders are published.

The Execution Summary message contains the instrument identifier, side, , worst price, total executed quantity, resting hidden quantity (if any) and match-time information of the incoming order.

For conveying the individual executions of the visible orders two template messages will be used for fully and partially executed orders.

The individual order execution messages should be used by participants for order book maintenance to ensure the correctness of the order book. The Execution Summary messages can be used by participants for fast trading decisions.

However, it should also be noted that, the Execution Summary message will **not** be published in the case a match is not triggered by an incoming order. It is illustrated by the following usecase.

The order execution messages will be sent whenever a visible order is **fully** or **partially** executed at its displayed price. Each **matchstep** will include a **product-wide day-unique identifier** of the trade, represented by the field *TrdMatchID*. This field will always have a value in the execution messages for a full or partial execution. The same unique identifier of the trade is made available to participants by the MCX T7 ETI.

If the incoming order has been partially executed, then the remaining quantity will be reported with an Order Add message after all associated individual executions have been provided.

Triggered Stop Market orders or Stop Limit orders are reported like incoming Market or Limit orders, respectively.

3.9 Trade Statistics

Instrument trade statistics such as opening, closing, daily low and high prices are available via the MCX T7 EOBI snapshot messages only. They are provided to participants for recovery purposes and are published included in the Instrument Summary message on the MCX T7 EOBI snapshot channel. By design, they are provided as a repeating group as part of the Instrument Summary message and are not cut off.

When subscribed to the MCX T7 EOBI incremental channel, participants can derive order book and trade statistics by combining the information received via the order and execution messages.

3.10 Product and Instrument States

In a Product State Change message, the product state can normally be found in the field *TradingSessionSubID*. Only for quiescent product states, the field *TradingSessionID* must be evaluated additionally to determine the actual product state.

A Halt state is additionally indicated by the field *TradSesStatus* containing the value "1 =Halted".

A Fast Market is reported with the same message type using the field *FastMarketIndicator* which can take the values “0 = No” or “1 = Yes”.

The instrument state is published with an InstrumentStateChange message and can be found directly in the field *SecurityTradingStatus*.

The status of the instrument (as opposed to the instrument state) distinguishes active, suspended and inactive instruments and is contained in the field *SecurityStatus*.

3.11 Heartbeats

Functional heartbeat messages, Heartbeat, are sent at a regular interval for less active products on the MCX T7 EOBI incremental channels. A functional heartbeat message provides the message sequence number last sent in the field *LastMsgSeqNumProcessed* to allow participants to identify potential gaps. Heartbeats will be sent out as of the product state “Start-Of-Day”.

Technical heartbeats will be provided on the specific ports assigned to technical heartbeat messages.

3.12 Recovery

Due to the unreliable nature of UDP multicast, UDP packets may be duplicated, delayed, missing, or arrive in an incorrect sequence. Participants can utilize the MCX T7 EOBI snapshot channel to obtain the corresponding lost information, i.e., rebuild the initial order book, determine trade statistics and instrument states. For recovery, participants should recover on a product level (i.e., for all instruments of one product), for following reason:

- The field *LastMsgSeqNumProcessed* in the snapshot cycle is given on product level, so in order to synchronize the MCX T7 EOBI snapshot channel and the MCX T7 EOBI incremental channel, participants should recover for all instruments in the product.

Participant Fail-Over

In the event of a packet loss on both (Live - Live) services of an MCX T7 EOBI channel, recovery on the participant side can be achieved by recovering the order book information via the MCX T7 EOBI snapshot channel.

The MCX T7 EOBI snapshot channel is synchronized with the MCX T7 EOBI incremental channel through the use of message sequence numbering. Participants should subscribe to the MCX T7 Order Book Snapshot channel while buffering incoming messages from the MCX T7 EOBI incremental channel. Any incoming message from the MCX T7 EOBI incremental channel with a *MsgSeqNum* higher than the value of the *LastMsgSeqNumProcessed* field received in the Product Summary snapshot message should be applied to the order books after the full product snapshot is processed.

Exchange failure

A failure of a MCX T7 EOBI service for a certain *PartitionID* (5948) always leads to a full restart of the respective service and can be detected on an EOBI channel by following characteristics:

- The *ApplSeqNum* in the Packet Header is reset to 1.
- The *MsgSeqNum* for each product or *MarketSegment* in the Message Header is reset to 1.

When a participant receives packets on a specific multicast address (either on service A or service B) with an unexpected (lesser or equal) packet header *ApplSeqNum* (usually 1), it is advised, that the participant rebuilds his order books from the new incremental message sequence or subscribes to the MCX T7 EOBI snapshot channel again.

Note that, because of the unreliable nature of the UDP protocol, packets may arrive out of sequence. An application might also see packets with an *AppSeqNum* greater or equal to the previous *AppSeqNum* for a specific fail-over period. Whenever an application detects an unexpected new (lesser or equal) *AppSeqNum* on a specific multicast address with a packet header *TransactTime* t_0 from a new sender, all packets from the old sender are expected to have a packet header *TransactTime* $t < t_0$.

In certain cases of a full restart of a MCX T7 EOBI service, participants must also wait for the first message after the restart to be certain that a restart was executed.

The field *AppSeqResetIndicator* is always set in the Packet Header of the first few incremental messages after a (re-) start.

3.13 Availability of Enhanced Order Book Service

The MCX T7 EOBI is available during the entire business day between product states “Start-Of-Day” and “Post-End-Of-Day”.

Table 4 below shows the information typically sent on the MCX T7 EOBI during each product state. The messages listed in the table should serve as a super-set of messages and inform participants on “what-to expect” during each product state. However, it does not state any deterministic behaviour and should only be used as a guideline. The actual message set could be a sub-set of the listed messages depending on market conditions.

Product State	Messages
Start-Of-Day	Product State Change, Instrument State Change, Product Summary, Instrument Summary (incl. Trade Statistics), Heartbeat
Pre-Trading	Product State Change, Instrument State Change, Order Mass Delete, Product Summary, Instrument Summary (incl. Trade Statistics), Heartbeat
Trading	Product State Change, Instrument State Change, Add Order, Modify Order, Modify Order Same Priority, Delete Order, Partial Order Execution, Full Order Execution, Execution Summary, Heartbeat, Product Summary, Instrument Summary (incl. Trade Statistics), Snapshot Order,
Post-Trading	Product State Change, Instrument State Change, Order Mass Delete, Product Summary, Instrument Summary (incl. Trade Statistics), Top Of Book, Heartbeat

Product State	Messages
End-Of-Day	Product State Change, Instrument State Change, Product Summary, Instrument Summary (incl. Trade Statistics), Top Of Book, Heartbeat
Post-End-Of-Day	-
Halt	Product State Change, Instrument State Change, Order Mass Delete, Product Summary, Instrument Summary (incl. Trade Statistics)
Holiday	Product State Change, Instrument State Change, Product Summary, Instrument Summary (incl. Trade Statistics), Heartbeat

Table 4 - Availability of Order Book Messages within Different Product States

Please note that the MCX T7 EOBI snapshot channels stop after migration of all products to “Post- End-Of-Day”.

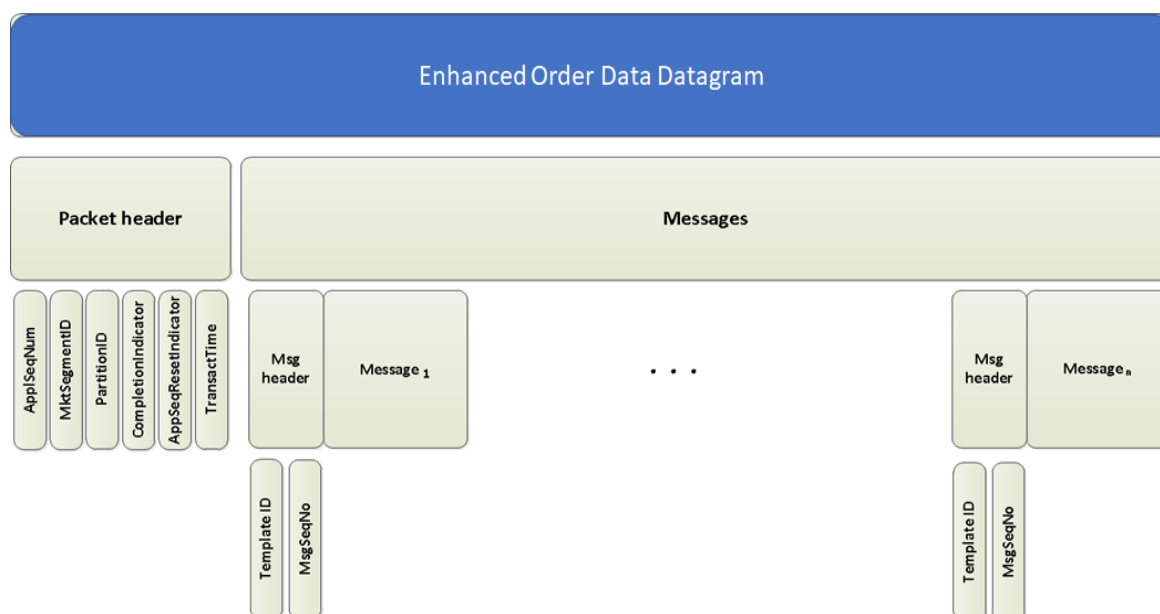
3.14 Message Formats

This chapter provides a global overview of the structure of datagram and message layouts and the data types used in these messages.

3.15 Datagram Structure

Each UDP datagram starts with a Packet Header followed by one or more public market data messages and is terminated on the product level boundary, meaning that a datagram contains not more than order book updates for one product.

The MCX T7 EOBI follows the following structure for the datagrams sent on the network:



Picture 4: Generic Datagram structure of MCX T7 EOBI

The Packet Header in each datagram contains information about

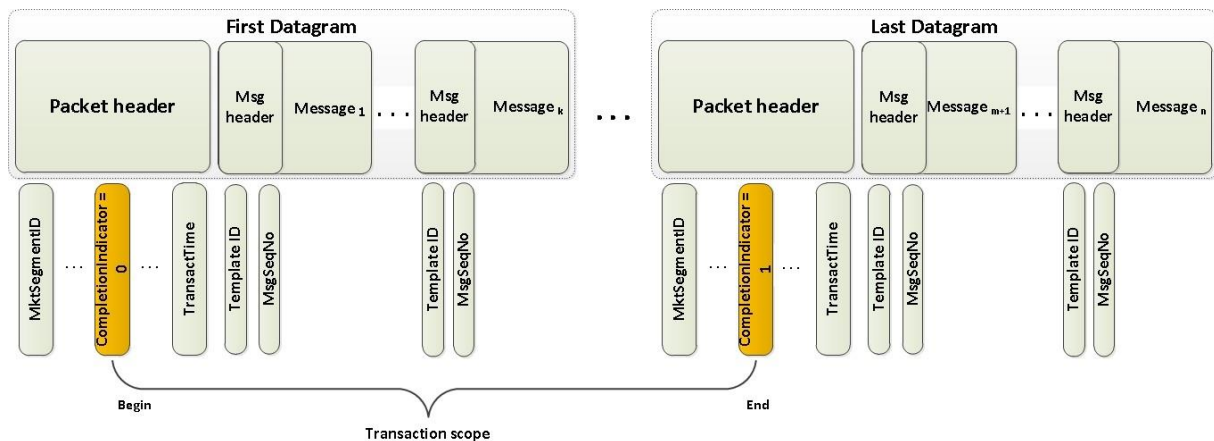
- The product and the partition ID of corresponding product,
- A contiguous packet sequence number,
- An indicator whether the **atomic unit of work** fits into one datagram,
- An indicator whether a fail-over has occurred, and
- When the packet has been sent out.

The product, *MarketSegmentID*, information can be used by participants for product filtering purposes.

The packet sequence numbers, *ApplSeqNum*, are contiguous and are incremented per MCX T7 EOB channel (service A and service B). They can be used by participants to detect gaps, duplicate and missing packets.

Please note, that EOB channels are not shared between different partitions.

Furthermore, the Packet Header provides information whether the atomic unit of work that was processed by the corresponding matching engine fits into one datagram or is spread over several datagrams. By design, a datagram will contain one atomic unit of work that was processed by the corresponding matching engine. However, if the resulting public market data of one atomic unit of work doesn't fit into one datagram due to datagram size restriction, then the resulting market data information is spread over several datagrams. In this case, as it is shown in the picture below, the completion flag, i.e., *CompletionIndicator*, in the first packet header of the first datagram is set to *Incomplete* (=0) and in the packet header of the last datagram is set to *Complete* (=1). As a result, participants are able to gather all market data information belonging together.



Picture 5: Transaction scope spread over several datagrams

When the public market data fits into one datagram, the completion indicator in the packet header will be set to *Complete* (=1).

The time when the datagram is sent out is provided by, *TransactTime*.

The functional structure of each MCX T7 EOB datagram will always be the same; a message header will specify the fixed layout of the message content by a *templateID*, followed by a message sequence number of the corresponding product. Message sequence numbers, *MsgSeqNum*, contained in the MCX T7 EOB incremental messages are incremented per product. Message sequence numbers for the MCX T7 EOB snapshot messages are incremented per snapshot cycle.

The repeating groups in incremental and snapshot messages are not cut off.

3.16 Incremental Messages

Incremental messages are sent according to the MCX T7 EOBi datagram structure as described above.

A message header will indicate the fixed layout of the message content, followed by the actual messages.

There is **no well-defined sending order** for the incremental messages. However, the *templateID* in the message header identifies each incremental message uniquely.

MCX T7 EOBi incremental messages will be sent as long as the MCX T7 EOBi service is available. The Heartbeat messages are repeated in the configured heartbeat interval in a single datagram by setting the message sequence number last sent to the *LastMsgSeqNumProcessed* field of the corresponding product. If the *LastMsgSeqNumProcessed* is not available, i.e., until the product state "Start-Of-Day", then it is set to "0".

As noted, if one atomic unit of work doesn't fit in one datagram, then the resulting market data information is spread over several datagrams. The completion flag will be used for this scenario.

Message	Template ID
Order Add	13100
Order Modify	13101
Order Modify Same Priority	13106
Order Delete	13102
Order Mass Delete	13103
Partial Order Execution	13105
Full Order Execution	13104
Execution Summary	13202
Top Of Book	13504
Product State Change	13300
Instrument State Change	13301
Heartbeat	13001
Trade Report	13201
Index Info	13604

Table 5 - MCX T7 Enhanced Order Book incremental messages

For order book maintenance, the order messages Order Add, Order Modify, Order Delete and Order Mass Delete will be provided along with the product and instrument state messages. Execution for orders will be published via Partial Order Execution and Full Order Execution messages for partially and fully matched orders. Additionally, an execution summary, Execution Summary, message will be provided for the mass execution scenarios.

Any update to the complex instruments will be provided via complex instrument messages. Auction information will be published as described in 4.8 - "Auctions" in detail.

Manually entered trades and reversed trades by MCX Market Supervision will be published by using Trade Report and Trade Reversal messages.

Cross Trade Announcements and Request for Quotes are disseminated by via the CrossRequest and the Quote Request messages. Request for Quotes and Cross Trade Announcements will be published via incremental messages only.

Functional Heartbeats will be published if there is no activity on a specific product.

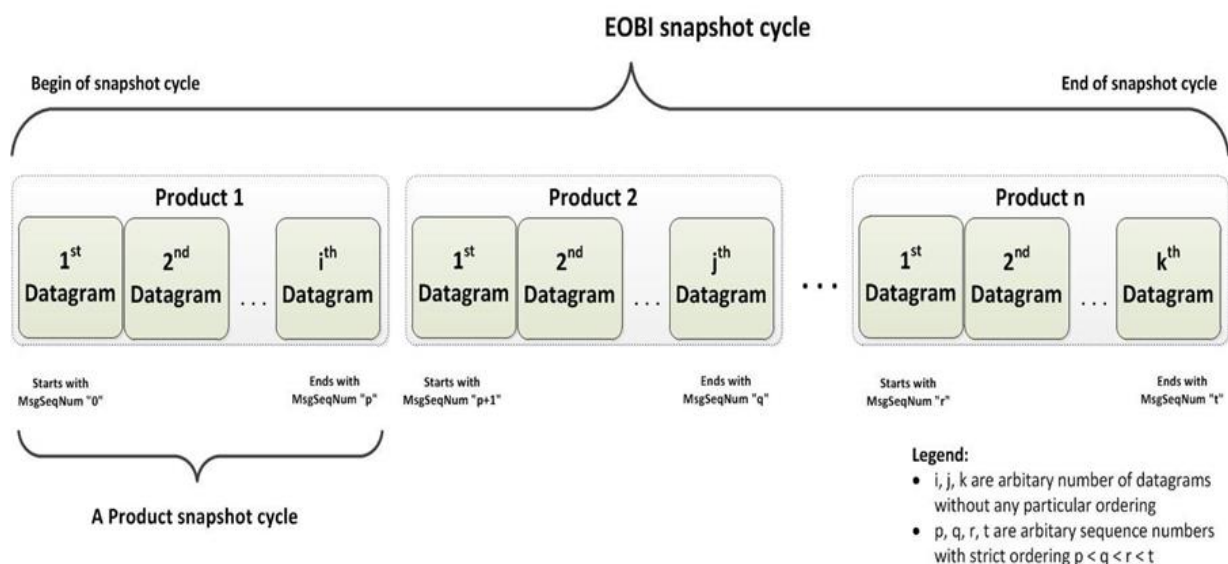
3.17 Snapshot Messages

By design, the snapshot messages are sent periodically and can be used by participants for recovery purposes, i.e. start-up processing or closing gaps in incremental messages. In contrast to MCX T7 EOBI incremental messages, MCX T7 EOBI snapshot messages will provide the trade statistics information at the time of sending. Furthermore, they contain the last message sequence number sent on the incremental feed, to provide a synchronization mechanism to participants for incremental and snapshots messages.

Like incremental messages, the snapshot messages will follow the MCX T7 EOBI - "Datagram Structure".

MCX T7 EOBI snapshot messages will be sent in product states between "Start-Of-Day" and "Post-End-Of-Day".

The picture below provides an overview of a typical **snapshot cycle**.



Picture 6: An overview of a snapshot cycle

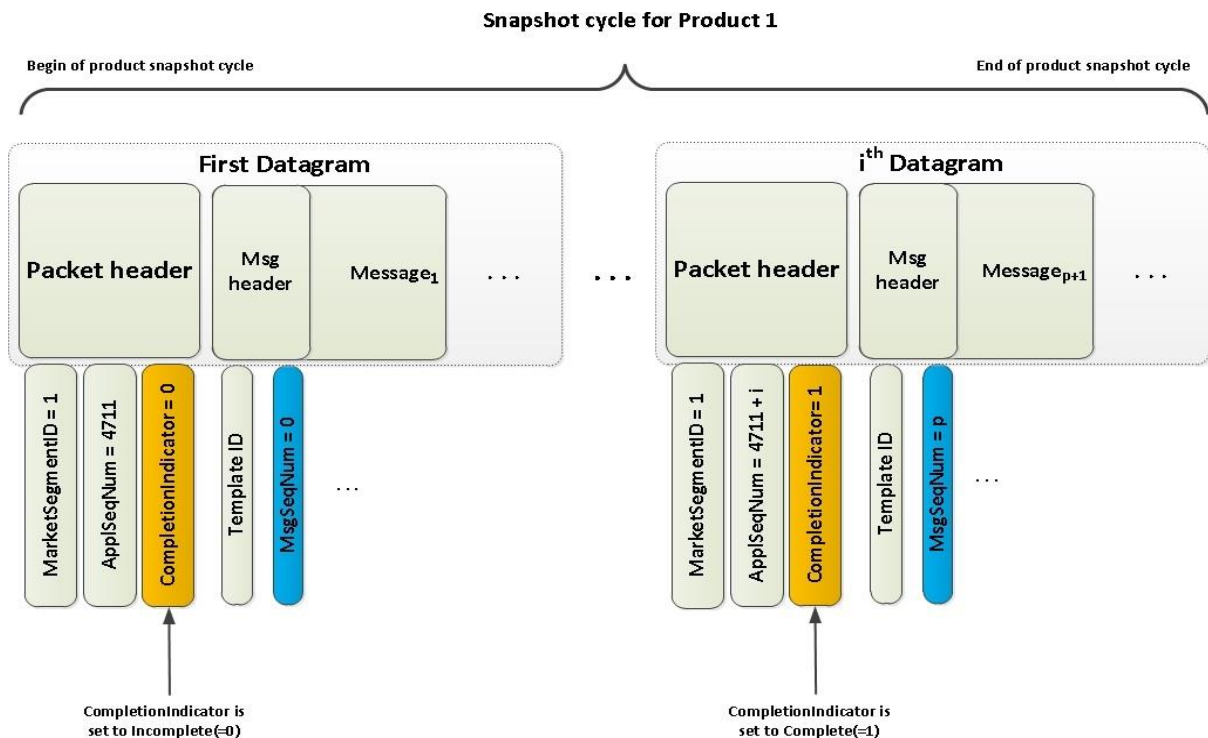
It is characterized by,

- The packet sequence numbers, *ApplSeqNum*, are contiguous and are incremented across products,
- The message sequence number, *MsgSeqNum*, of the first message in the first datagram of a new snapshot cycle is set to zero (=0),
- The message sequence number, *MsgSeqNum*, within the same snapshot cycle is incremented for each message across all messages and all products,
- The *CompletionIndicator* in the last datagram of a product snapshot cycle is set to *Complete*(=1) to inform about the end of a product snapshot cycle.

That implies, a full snapshot cycle on MCX T7 EOBI snapshot feed comprises of multiple product snapshot cycles. In order to assist an easy identification of a product snapshot boundary, the *CompletionIndicator* is set to *Complete*(=1) in the

last datagram of a product. Each snapshot cycle starts by re-setting the message sequence number, *MsgSeqNum*, to zero(=0) for the first message in the first datagram.

The following picture further outlines **product snapshot cycle** for the *Product1* from the picture above.



Each message header containing the *templateID* of a message within a snapshot cycle will specify the message content. Two summary messages are introduced to reduce the total size of snapshot messages in a snapshot cycle by avoiding redundant information:

- A Product Summary containing the last message sequence number of the last message sent on the incremental feed and trading state information, and
- An Instrument Summary for each instrument of the product including instrument state information and trade statistics such as last trade price and volume, daily low and high prices, opening prices etc. Additionally, the number of visible orders in the current product's snapshot cycle is provided to participants in advance.

The last message sequence number, *LastMsgSeqNumProcessed*, in the product summary message denotes the last message sent on the incremental feed, i.e., it provides a link between incremental and snapshot feed.

A snapshot cycle might contain order book information for multiple products. The following describes the snapshot cycle for one product.

A product has multiple instruments. The Product Summary will be given once, as it includes attributes that are identical for all instruments. However, it can include multiple InstrumentSummary messages, each followed by the individual orders for that instrument.

As it shown in picture below, a **snapshot cycle of a product** will always start with a product summary followed by an instrument summary followed by all visible orders of the corresponding instrument and so on. Logically, the whole process is repeated for all instruments of a product.



Picture 8: A snapshot cycle of a product

Finally, as snapshot cycle of product is terminated on the product level boundary, i.e., *CompletionIndicator* is set to *Complete(=1)*, the next Product Summary message implicitly defines the start of a snapshot cycle for the next product, inherently defining the product level boundary. All messages within a product level boundary are self-contained.

Order messages within a snapshot cycle will be sent in a zig-zag manner as described in 4.1 -“Building the Order Book“. All subsequent products follow a similar pattern, forming a snapshot cycle.

MCX T7 EOB snapshot messages will contain order book information about the intra-day created complex instruments as well, even if there is no trading activity in that complex instrument.

Please note that, during Auctions the snapshot messages contain either Auction Best Bid-Offer or Auction Clearing Price messages instead of the order messages, i.e., visible orders aren’t published during Auctions via snapshot messages.

Additionally, the Top Of Book messages will be published starting from post trading state until end of day trading state to provide participants with last available instrument’s BBO information.

3.18 Data Types

The following table provides an overview of the data types used in the fixed-length binary encoded messages sent out by the MCX T7 EOB. These data types will be indicated for each field in the Chapter 8 - “Message Layout“, which covers the message layouts.

Data Type	Description	No Value
signed int.	little endian byte order supported are 1, 2, 4 and 8-byte, signed integers the most significant bit contains the sign.	1 byte signed int: 0x80 2 byte signed int: 0x8000 4 byte signed int: 0x80000000 8 byte signed int: 0x8000000000000000
unsigned int.	little endian byte order supported are 1, 2, 4 and 8-byte unsigned integer.	1 byte unsigned int: 0xFF 2 byte unsigned int: 0xFFFF 4 byte unsigned int: 0xFFFFFFFF 8 byte unsigned int: 0xFFFFFFFFFFFFFFFF
PriceType	Price in integer format including 8 decimals. For certain asset classes, prices may have negative values.	see 8 byte signed int.
QuantityType	Quantity in integer format including 4 decimals.	see 8 byte signed int.
Counter	Contains a record or message counter.	see 4 byte signed int.
UTCTimestamp	Date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00UTC on January 1 st , 1970).	see 8 byte unsigned int.

Table 6 - Data types on the MCX T7 EOB

4 Functional Specification

4.1 Overview of Supported Message Types and Reserve fields

The mapping of the MCX Reserve fields to the FIX tags is as follows:

Reserve Fields

EOBI Field	Relevant FIX TagID	Description
Reserve1	5979	Gateway request in timestamp
Reserve2	21026	Previous order priority timestamp.
Reserve3	2445	Matching Engine-In timestamp
Reserve4	21008	Priority timestamp (new)

The following message formats are based on:

General

EOBI Message	TemplateID (28500)	FIX Message	MsgType (35)
Packet Header	13003	MarketDataReport	U20
Heartbeat	13001	Heartbeat	0

Trade Data

EOBI Message	TemplateID (28500)	FIX Message	MsgType (35)
Execution Summary	13202	MarketDataTrade	U22
Index Information	13604	MarketDataTrade	U22
Trade Report	13201	MarketDataTrade	U22

Order Data

EOBI Message	TemplateID (28500)	FIX Message	MsgType (35)
Order Add	13100	MarketDataOrder	U21
Top of Book	13504	MarketDataInstrument	U23
Order Modify	13101	MarketDataOrder	U21
Order Modify Same Priority	13106	MarketDataOrder	U21
Order Delete	13102	MarketDataOrder	U21
Order Mass Delete	13103	MarketDataOrder	U21
Partial Order Execution	13105	MarketDataOrder	U21
Full Order Execution	13104	MarketDataOrder	U21

State Change

EOBI Message	TemplateID (28500)	FIX Message	MsgType (35)
Product State Change	13300	TradingSessionStatus	h
Mass Instrument State Change	13302	SecurityMassStatus	CO
Instrument State Change	13301	SecurityStatus	f

Snapshot

EOBI Message	TemplateID (28500)	FIX Message	MsgType (35)
Product Summary	13600	MarketDataInstrument	U23
Instrument Summary	13601	MarketDataInstrument	U23
Snapshot Order	13602	MarketDataOrder	U21

4.2 Packet Header = 13003

The Packet Header is a technical header which is delivered in every UDP-datagram, and is used for identification of datagrams. The Packet Header will be published on a multicast channel basis, with each packet containing information for one product only, recognizable by the field MarketSegmentID. Whenever there is an amount of information that doesn't fit in one datagram, the field CompletionIndicator will be set to 'Incomplete'. A CompletionIndicator field set to 'Incomplete' implies that another (new) datagram will follow, containing the remaining data. This will be applied to the incremental messages only. Every partition stamps the outgoing datagrams with a sequence number: ApplSeqNum and a sending time: TransactTime. It also includes the ApplSeqResetIndicator field that can be set incase of market data fail-over and/or a market data restart.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13003 (Market-DataReport, MsgType = U20)
34	MsgSeqNum	U	4	4	unsigned int	not used
<Message Body>						
1181	ApplSeqNum	Y	4	8	unsigned int	Message sequence number is contiguous and is incremented across products.
1300	MarketSegmentID	Y	4	12	signed int	Product identifier.
5948	PartitionID	Y	1	16	unsigned int	Grouping of MCX T7 products. Belongs to the scope of Service Availability.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
6228	CompletionIndicator	Y	1	17	unsigned int	Indicated whether an unit of works fitsinto a single datagram for incremental messages.	
						Value	Description
						0	Incomplete
						1	Complete
28841	ApplSeqResetIndicator	Y	1	18	unsigned int		
						Value	Description
						0	Incomplete
						1	Complete
25020	Pad5	U	5	19	Fixed String	not used	
60	TransactTime	Y	8	24	UTCTimestamp	Time in nanoseconds when market data feed handler writes packet on the wire.	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U20	3	Fixed String	U20 = Market Data Report
28827	MDReportEvent	0	1	unsigned int	0 = Scope Definition.

4.3 Heartbeat = 13001

A functional Heartbeat message will be published regularly per product when there is no activity on the MCX T7 Enhanced Order Book Interface incremental channel. The functional Heartbeat message will contain the last processed message sequence number, enabling participants to check for missed or lost packets.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13001 (Heartbeat, MsgType = 0)
34	MsgSeqNum	U	4	4	unsigned int	not used
<Message Body>						
369	LastMsgSeqNum-Processed	Y	4	8	unsigned int	Last Message Sequence number that was processed, regardless of message type.
25019	Pad4	U	4	12	Fixed String	not used

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	0	3	Fixed String	0 = Heartbeat

4.4 Execution Summary = 13202

Whenever an incoming order is executed, an *Execution Summary* message will be published, containing information on the execution of the incoming order. The *Execution Summary* message only contains information for the initial instrument (security), that was specified by the incoming order, i.e. any synthetic matches/changes can not be derived from the summary message. The *Execution Summary* message may be used for fast trading decisions. In fact, to be absolutely sure the order book is correct, participants should always process the execution messages following the *Execution Summary* message.

The fields in the *Execution Summary* message provide information on the instrument specified in the incoming order, transaction time, the side of the incoming order, an indicator for a synthetic match, the quantity that was executed (of the specified instrument) in the fill, and the worst price of the fill, represented by the fields *SecurityID*, *ExecID*, *AggressorSide*, *TradeCondition*, *LastQty*, *RestingHiddenQty* and *LastPx* respectively.

The *RestingHiddenQty* in the context of an execution (of the specified instrument) would refer to the resting hidden quantity included in the sum of *LastQty* and *RestingCxlQty*. It is set to zero, if no such quantity is involved.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description						
<MessageHeader>												
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.						
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13202 (Market-DataTrade, MsgType = U22)						
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.						
<Message Body>												
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.						
2445	Reserve3	N	8	16	UTCTimestamp	not used						
5979	Reserve1	N	8	24	UTCTimestamp	not used						
17	ExecID	Y	8	32	UTCTimestamp	Transaction time stamp						
32	LastQty	Y	8	40	QuantityType	Total executed matched quantity of this match event.						
2446	AggressorSide	Y	1	48	unsigned int	<table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Triggered by the buy side</td></tr><tr><td>2</td><td>Triggered by the sell side</td></tr></table>	Value	Description	1	Triggered by the buy side	2	Triggered by the sell side
Value	Description											
1	Triggered by the buy side											
2	Triggered by the sell side											
25016	Pad1	U	1	49	Fixed String	not used						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
277	TradeCondition	N	2	50	unsigned int	not used
25019	Pad4	U	4	52	Fixed String	not used
31	LastPx	Y	8	56	PriceType	Worst price of this match.
28868	RestingHiddenQty	N	8	64	QuantityType	Quantity of executed and/or cancelled passive orders that were not displayed to the market. Set to zero, if no such quantity is involved.
28869	RestingCxlQty	Y	8	72	QuantityType	Total cancelled (deleted) quantity due to Self Match Prevention (SMP) of this match event. This quantity is not part of LastQty which could even be 0 in certain cases.

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U22	3	Fixed String	U22 = Market Data Trade
28842	MarketDataType	12	1	unsigned int	12 = Match Event See also MCX T7 EOBI Schema (XSD) file.
279	MDUpdateAction	0	1	unsigned int	0 = New Type of Market Data update action.
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.5 Order Add = 13100

An Order Add message will be published for each new order that was entered in the order book.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13100 (Market-DataOrder, MsgType = U21)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description						
60	Transaction Time	N	8	8	UTCTimestamp	Transaction timestamp in nano-seconds						
48	SecurityID	Y	8	16	signed int	Unique instrument identifier.						
<OrderDetails>												
21008	Reseve2	Y	8	24	UTCTimestamp	Not used.						
1138	DisplayQty	Y	8	32	QuantityType	Quantity						
54	Side	Y	1	40	unsigned int	Side of the order. <table><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Buy</td></tr><tr><td>2</td><td>Sell</td></tr></table>	Value	Description	1	Buy	2	Sell
Value	Description											
1	Buy											
2	Sell											
40	OrdType	N	1	41	unsigned int	Used for cash market instruments only. 1 = Market Order Used for cash market instruments only. <table><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Market Order</td></tr></table>	Value	Description	1	Market Order		
Value	Description											
1	Market Order											
25021	Pad6	U	6	42	Fixed String	not used						
44	Price	N	8	48	PriceType	Price.						

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	1	1	unsigned int	1 = Order Book Maintenance
279	MDUpdateAction	0	1	unsigned int	0 = New Type of Market Data update action.
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.6 Top of Book = 13504

For derivatives market the Top of Book messages will be published via incremental and snapshot messages starting from post trading state until end of day trading state to provide the BBO instrument's information.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13504 (MarketDataInstrument, MsgType = U23)

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						
60	TransactTime	Y	8	8	UTCTimestamp	Transaction timestamp in nano-seconds
48	SecurityID	Y	8	16	signed int	Unique instrument identifier.
132	BidPx	N	8	24	PriceType	Bid price/rate.
133	OfferPx	N	8	32	PriceType	Offer price/rate.
134	BidSize	N	8	40	QuantityType	Quantity of bid.
135	OfferSize	N	8	48	QuantityType	Quantity of offer.
2449	NumberOfBuyOrders	N	2	56	unsigned int	Number of bid orders.
2450	NumberOfSellOrders	N	2	58	unsigned int	Number of offer orders.
25019	Pad4	U	4	60	Fixed String	not used

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U23	3	Fixed String	U23 = Market Data Instrument
28842	MarketDataType	13	1	unsigned int	13 = Top Of Book See also MCX T7 EOBI Schema (XSD) file.
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.7 Order Modify = 13101

An Order Modify message will be published, if an existing order in the book is modified, whereby the new parameters of the order might cause a change in time priority. If an order is modified to another price, or if the quantity of this order is increased, the time priority of the order will change.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13101 (Market-DataOrder, MsgType = U21)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						
60	Transaction Time	Y	8	8	UTCTimestamp	Transaction timestamp in nano-seconds
21026	Reserve2	Y	8	16	UTCTimestamp	Not used

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
28855	PrevPrice	N	8	24	PriceType	Previous order price.	
28867	PrevDisplayQty	Y	8	32	QuantityType	Previous display quantity	
48	SecurityID	Y	8	40	signed int	Unique instrument identifier.	
<OrderDetails>							
21008	Reserve4	Y	8	48	UTCTimestamp	Not used.	
1138	DisplayQty	Y	8	56	QuantityType	Quantity.	
54	Side	Y	1	64	unsigned int	Side of the order.	
						Value	Description
						1	Buy
						2	Sell
40	OrdType	N	1	65	unsigned int	Used for cash market instruments on-ly.	
						1 = Market Order Used for cash market instruments only.	
						Value	Description
						1	Market Order
25021	Pad6	U	6	66	Fixed String	not used	
44	Price	N	8	72	PriceType	Price.	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	1	1	unsigned int	1 = Order Book Maintenance See also MCX T7 EOBI Schema (XSD) file.
279	MDUpdateAction	1	1	unsigned int	1 = Change
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.8 Order Modify Same Priority = 13106

An Order Modify Same Priority message will be published, if the time priority of an existing order is not changed.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13106 (Market-DataOrder, MsgType = U21)

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.	
<Message Body>							
21002	Reserve1	Y	8	8	UTCTimestamp		
60	TransactTime	Y	8	16	UTCTimestamp	Transaction timestamp in nano-seconds	
28867	PrevDisplayQty	Y	8	24	QuantityType	Previous display quantity	
48	SecurityID	Y	8	32	signed int	Unique instrument identifier.	
<OrderDetails>							
21008	Reserve4	Y	8	40	UTCTimestamp	Not used	
1138	DisplayQty	Y	8	48	QuantityType	Quantity.	
54	Side	Y	1	56	unsigned int	Side of the order.	
						Value	Description
						1	Buy
						2	Sell
40	OrdType	N	1	57	unsigned int	Used for cash market instruments on-ly. 1 = Market Order Used for cash market instruments only.	
						Value	Description
						1	Market Order
25021	Pad6	U	6	58	Fixed String	not used	
44	Price	N	8	64	PriceType	Price.	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	1	1	unsigned int	1 = Order Book Maintenance
279	MDUpdateAction	1	1	unsigned int	1 = Change
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.9 Order Delete = 13102

Whenever an existing order is deleted from the order book, an Order Delete message will be published. The Order Delete message will contain all necessary fields needed to delete the correct order; SecurityID, Side. For convenience, the order delete message will also contain the former displayed quantity and the former price.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						

9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.	
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13102 (MarketDataOrder, MsgType = U21)	
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.	
<Message Body>							
21002	reserve1	N	8	8	UTCTimestamp		
60	TransactTime	Y	8	16	UTCTimestamp	Transaction timestamp in nano-seconds	
48	SecurityID	Y	8	24	signed int	Unique instrument identifier.	
<OrderDetails>							
21008	Reserve2	Y	8	32	UTCTimestamp	Not used.	
1138	DisplayQty	Y	8	40	QuantityType	Quantity.	
54	Side	Y	1	48	unsigned int	Side of the order.	
						Value	Description
						1	Buy
						2	Sell
40	OrdType	N	1	49	unsigned int	Used for cash market instruments on-ly.	
						1 = Market Order Used for cash market instruments only.	
						Value	Description
						1	Market Order
25021	Pad6	U	6	50	Fixed String	not used	
44	Price	N	8	56	PriceType	Price.	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	1	1	unsigned int	1 = Order Book Maintenance
279	MDUpdateAction	2	1	unsigned int	2 = Delete Type of Market Data update action.
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.10 Order Mass Delete = 13103

An Order Mass Delete message will be published when the order book is expected to be emptied. The message contains the instrument identifier indicating which order book has to be fully deleted.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13103 (Market-DataOrder, MsgType = U21)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.
60	TransactTime	Y	8	16	UTCTimestamp	Transaction timestamp in nano-seconds

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	1	1	unsigned int	1 = Order Book Maintenance See also MCX T7 EOB Schema (XSD) file.
279	MDUpdateAction	2	1	unsigned int	2 = Delete
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.11 Partial Order Execution = 13105

Whenever a visible order is partially executed at its displayed price, a Partial Order Execution message will be published, containing the execution information; instrument identifier, price and executed quantity of the executed passive order as well as the match identifier. The remaining quantity in the order book for this order must be calculated by subtracting the executed quantity in the Partial Order Execution message from the initial quantity in the order book.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13105 (Market-DataOrder, MsgType = U21)

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.	
<Message Body>							
54	Side	Y	1	8	unsigned int	Side of the order.	
						Value	Description
						1	Buy
						2	Sell
40	OrdType	N	1	9	unsigned int	Used for cash market instruments on-ly. 1 = Market Order Used for cash market instruments only.	
						Value	Description
						1	Market Order
2667	AlgorithmicTrade-Indicator	N	1	10	unsigned int	A trade is flagged as <i>algorithmic</i> , if atleast one of the matched orders was submitted by a trading algorithm. Applicable for cash market instruments only.	
						Value	Description
						1	Algorithmic Trade
25016	Pad1	U	1	11	Fixed String	not used	
880	TrdMatchID	Y	4	12	unsigned int	Unique identifier for each price level (match step) of a match event; it isused for public trade reporting.	
44	Price	N	8	16	PriceType	The price at which the order enteredthe book. Typically it is equal to Last-Px except during auction uncrossing.	
60	Transaction Time	Y	8	24	UTCTimestamp	Transaction timestamp in nano-seconds	
48	SecurityID	Y	8	32	signed int	Unique instrument identifier.	
32	LastQty	Y	8	40	QuantityType	Quantity executed in this fill.	
31	LastPx	Y	8	48	PriceType	The price at which the order was matched.	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	2	1	unsigned int	2 = Order Book Execution

279	MDUpdateAction	1	1	unsigned int	1 = Change
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.12 Full Order Execution = 13104

Whenever a visible order is fully executed at its displayed price, a Full Order Execution message will be published, containing the execution information; instrument identifier, price and executed quantity of the executed passive order and the match identifier. As this order is executed in full, it has to be deleted from the order book.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description						
<MessageHeader>												
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.						
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13104 (Market-DataOrder, MsgType = U21)						
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.						
<Message Body>												
54	Side	Y	1	8	unsigned int	Side of the order. <table><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Buy</td></tr><tr><td>2</td><td>Sell</td></tr></table>	Value	Description	1	Buy	2	Sell
Value	Description											
1	Buy											
2	Sell											
40	OrdType	N	1	9	unsigned int	Used for cash market instruments on-ly. 1 = Market Order Used for cashmarket instruments only. <table><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Market Order</td></tr></table>	Value	Description	1	Market Order		
Value	Description											
1	Market Order											
2667	AlgorithmicTrade-Indicator	N	1	10	unsigned int	A trade is flagged as <i>algorithmic</i> , if atleast one of the matched orders wassubmitted by a trading algorithm. Ap- plicable for cash market instruments only. <table><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Algorithmic Trade</td></tr></table>	Value	Description	1	Algorithmic Trade		
Value	Description											
1	Algorithmic Trade											
25016	Pad1	U	1	11	Fixed String	not used						
880	TrdMatchID	Y	4	12	unsigned int	Unique identifier for each price level (match step) of a match event; it is used for public trade reporting.						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
44	Price	N	8	16	PriceType	The price at which the order entered the book. Typically it is equal to Last-Px except during auction uncrossing.
60	Transaction Time	Y	8	24	UTCTimestamp	Transaction timestamp in nano-seconds
48	SecurityID	Y	8	32	signed int	Unique instrument identifier.
32	LastQty	Y	8	40	QuantityType	Quantity executed in this fill.
31	LastPx	Y	8	48	PriceType	The price at which the order was matched.

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	2	1	unsigned int	2 = Order Book Execution
279	MDUpdateAction	1	1	unsigned int	1 = Change
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.13 Product State Change = 13300

The Product State Change message provides updates on the trading state for (all instruments in) a particular product.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOB message layout. Value: 13300 (Trading-SessionStatus, MsgType = h)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
336	TradingSessionID	Y	1	8	unsigned int	Product state information.	
						Value	Description
						1	Day
						3	Morning
						5	Evening
						6	After Hours
						7	Holiday
625	TradingSessionSubID	Y	1	9	unsigned int	Product state information.	
						Value	Description
						1	Pre Trading
						3	Continuous
						4	Closing
						5	Post Trading
						7	Quiescent
340	TradSesStatus	Y	1	10	unsigned int	Product state information.	
						Value	Description
						1	Halted
						2	Open
						3	Closed
2705	MarketCondition	N	1	11	unsigned int	Indicator for stressed market conditions.	
						Value	Description
						0	Normal
						1	Stressed
2447	FastMarketIndicator	Y	1	12	unsigned int	Indicates if product is in state "Fast Market". This indicator refers to a product but is provided on instrument level.	
						Value	Description
						0	No
						1	Yes
25018	Pad3	U	3	13	Fixed String	not used	
60	TransactTime	Y	8	16	UTCTimestamp	Transaction timestamp in nano-seconds	

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	h	3	Fixed String	h = Trading Session Status
1368	TradSesEvent	3	1	unsigned int	3 = Status Change

ex

4.14 Mass Instrument State Change = 13302

The Mass Instrument State Change message provides the state information for all instruments of a certain instrument type or *InstrumentScopeProductComplex* (1544) within a product.

Where not all indicated instruments are affected by the new state, the exception list is populated with one entry for each such instrument.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
<MessageHeader>							
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.	
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13302 (SecurityMassStatus, MsgType = CO)	
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.	
<Message Body>							
1544	InstrumentScope-ProductComplex	Y	1	8	unsigned int	Instrument type of affected instruments	
						Value	Description
						1	Simple Instrument
						5	Futures Spread
30965	SecurityMassStatus	Y	1	9	unsigned int	The instrument status of all affected instruments.	
						Value	Description
						1	Active
						2	Inactive
						4	Expired
9	Suspended						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description												
1679	SecurityMassTrading-Status	Y	1	10	unsigned int	The instrument trading state of all affected instruments. <table><tr><td>Value</td><td>Description</td></tr><tr><td>2</td><td>Trading Halt</td></tr><tr><td>200</td><td>Closed</td></tr><tr><td>201</td><td>Restricted</td></tr><tr><td>202</td><td>Book</td></tr><tr><td>203</td><td>Continuous</td></tr></table>	Value	Description	2	Trading Halt	200	Closed	201	Restricted	202	Book	203	Continuous
Value	Description																	
2	Trading Halt																	
200	Closed																	
201	Restricted																	
202	Book																	
203	Continuous																	
28894	MassMarketCondition	Y	1	11	unsigned int	Not used. This is set to 0 - Normal												
2447	FastMarketIndicator	Y	1	12	unsigned int	Not used. Set to 0 by default												
1680	SecurityMassTrading-Event	N	1	13	unsigned int	Identifies an event related to a <i>SecurityMassTradingStatus (1679)</i>. Used for cash market instruments only. <table><tr><td>Value</td><td>Description</td></tr><tr><td>10</td><td>Price volatility, auction is extended</td></tr><tr><td>11</td><td>Price volatility, auction is extended again</td></tr></table>	Value	Description	10	Price volatility, auction is extended	11	Price volatility, auction is extended again						
Value	Description																	
10	Price volatility, auction is extended																	
11	Price volatility, auction is extended again																	
35155	MassSoldOutIndicator	N	1	14	unsigned int	Not used. Only applicable for trading model Continuous Auction Issuer for cash market products.												
25016	Pad1	U	1	15	Fixed String	not used												
60	TransactTime	Y	8	16	UTCTimestamp	Transaction timestamp in nano-seconds												
893	LastFragment	Y	1	24	unsigned int	<table><tr><td>Value</td><td>Description</td></tr></table>	Value	Description										
Value	Description																	

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description												
						<table><tr><td>0</td><td>N</td></tr><tr><td>1</td><td>Y</td></tr></table> Indicates whether this message is the last in a sequence of messages that together convey a joint exception list of SecMassStatGrp. All messages up to the last with <i>LastFragment</i> = Y share the same root level content and an application first needs to combine all single exception lists before the MassState Change message could be applied with the fully joint exception list. N = Not Last Message Y = Last Message	0	N	1	Y								
0	N																	
1	Y																	
146	NoRelatedSym	Y	1	25	Counter	Specifies the number of following instrument state exceptions.												
25021	Pad6	U	6	26	Fixed String	not used												
<SecMassStatGrp>						Variable size array, Record counter: NoRelatedSym												
48	>SecurityID	Y	8	32	signed int	Unique instrument identifier.												
965	>SecurityStatus	Y	1	40	unsigned int	See <i>Instrument State Change</i>. <table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Active</td></tr><tr><td>2</td><td>Inactive</td></tr><tr><td>4</td><td>Expired</td></tr><tr><td>9</td><td>Suspended</td></tr></table>	Value	Description	1	Active	2	Inactive	4	Expired	9	Suspended		
Value	Description																	
1	Active																	
2	Inactive																	
4	Expired																	
9	Suspended																	
326	>SecurityTrading-Status	Y	1	41	unsigned int	See <i>Instrument State Change</i>. <table><tr><th>Value</th><th>Description</th></tr><tr><td>2</td><td>Trading Halt</td></tr><tr><td>200</td><td>Closed</td></tr><tr><td>201</td><td>Restricted</td></tr><tr><td>202</td><td>Book</td></tr><tr><td>203</td><td>Continuous</td></tr></table>	Value	Description	2	Trading Halt	200	Closed	201	Restricted	202	Book	203	Continuous
Value	Description																	
2	Trading Halt																	
200	Closed																	
201	Restricted																	
202	Book																	
203	Continuous																	
2705	>MarketCondition	Y	1	42	unsigned int	Not used. Set to 0- Normal												
1174	>SecurityTrading-Event	N	1	43	unsigned int	Not used.												

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
25155	>SoldOutIndicator	N	1	44	unsigned int	Not used.
25018	>Pad3	U	3	45	Fixed String	Not used

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	CO	3	Fixed String	CO = Security Mass Status
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.15 Instrument State Change = 13301

The Instrument State Change message provides state information for a single instrument. Furthermore, it informs participants about intra-day expiration of instruments.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description										
<MessageHeader>																
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.										
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13301 (Security-Status, MsgType = f)										
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all messagetypes.										
<Message Body>																
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.										
965	SecurityStatus	Y	1	16	unsigned int	<table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Active</td></tr><tr><td>2</td><td>Inactive</td></tr><tr><td>4</td><td>Expired</td></tr><tr><td>9</td><td>Suspended</td></tr></table>	Value	Description	1	Active	2	Inactive	4	Expired	9	Suspended
						Value	Description									
						1	Active									
						2	Inactive									
						4	Expired									
9	Suspended															
326	SecurityTradingStatus	Y	1	17	unsigned int	<table><tr><th>Value</th><th>Description</th></tr><tr><td>2</td><td>Trading Halt</td></tr><tr><td>200</td><td>Closed</td></tr><tr><td>201</td><td>Restricted</td></tr><tr><td>202</td><td>Book</td></tr></table>	Value	Description	2	Trading Halt	200	Closed	201	Restricted	202	Book
						Value	Description									
						2	Trading Halt									
						200	Closed									
						201	Restricted									
202	Book															

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
						203 Continuous
2705	MarketCondition	Y	1	18	unsigned int	Not used
2447	FastMarketIndicator	Y	1	19	unsigned int	Not used
1174	SecurityTradingEvent	N	1	20	unsigned int	Not used
25155	SoldOutIndicator	N	1	21	unsigned int	Not used
25017	Pad2	U	2	22	Fixed String	not used
60	TransactTime	Y	8	24	UTCTimestamp	Transaction timestamp in nano-seconds

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	f	3	Fixed String	f = Security Status
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.16 Product Summary = 13600

A Product Summary message will be published once each snapshot cycle, and will contain attributes that are equal for all instruments that belong to that product.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13600 (Market-DataInstrument, MsgType = U23)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product.
<Message Body>						

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description														
369	LastMsgSeqNum-Processed	Y	4	8	unsigned int	Last Message Sequence number that was processed, regardless of message type.														
336	TradingSessionID	N	1	12	unsigned int	<table><tr><td colspan="2">Product state information.</td></tr><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Day</td></tr><tr><td>3</td><td>Morning</td></tr><tr><td>5</td><td>Evening</td></tr><tr><td>6</td><td>After Hours</td></tr><tr><td>7</td><td>Holiday</td></tr></table>	Product state information.		Value	Description	1	Day	3	Morning	5	Evening	6	After Hours	7	Holiday
Product state information.																				
Value	Description																			
1	Day																			
3	Morning																			
5	Evening																			
6	After Hours																			
7	Holiday																			
625	TradingSessionSubID	N	1	13	unsigned int	<table><tr><td colspan="2">Product state information.</td></tr><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Pre Trading</td></tr><tr><td>3</td><td>Continuous</td></tr><tr><td>4</td><td>Closing</td></tr><tr><td>5</td><td>Post Trading</td></tr><tr><td>7</td><td>Quiescent</td></tr></table>	Product state information.		Value	Description	1	Pre Trading	3	Continuous	4	Closing	5	Post Trading	7	Quiescent
Product state information.																				
Value	Description																			
1	Pre Trading																			
3	Continuous																			
4	Closing																			
5	Post Trading																			
7	Quiescent																			
340	TradSesStatus	N	1	14	unsigned int	<table><tr><td colspan="2">Product state information.</td></tr><tr><td>Value</td><td>Description</td></tr><tr><td>1</td><td>Halted</td></tr><tr><td>2</td><td>Open</td></tr><tr><td>3</td><td>Closed</td></tr><tr><td colspan="2"></td></tr></table>	Product state information.		Value	Description	1	Halted	2	Open	3	Closed				
Product state information.																				
Value	Description																			
1	Halted																			
2	Open																			
3	Closed																			
2705	MarketCondition	N	1	15	unsigned int	Not used.														
2447	FastMarketIndicator	Y	1	16	unsigned int	Not used.														
25022	Pad7	U	7	17	Fixed String	not used														

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U23	3	Fixed String	U23 = Market Data Instrument
28842	MarketDataType	9	1	unsigned int	9 = Market Segment Snapshot

4.17 Instrument Summary = 13601

An Instrument Summary message will be published for each instrument in one snapshot cycle on the MCX T7 Enhanced Order Book Interface snapshot channel, and will contain instrument state information and trade statistics for one instrument. Note that one product can have multiple instruments. The repeating group MDEntryGrp, instrument's trade statistics, are not cut off by design.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description														
<MessageHeader>																				
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.														
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13601 (Market-DataInstrument, MsgType = U23)														
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product.														
<Message Body>																				
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.														
779	LastUpdateTime	Y	8	16	UTCTimestamp	Last update time of the corresponding order book.														
21001	TrdRegTSExecution-Time	N	8	24	UTCTimestamp	Last Trade Time														
68	TotNoOrders	Y	2	32	Counter	Corresponding number of orders for this instrument.														
965	SecurityStatus	Y	1	34	unsigned int	<table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Active</td></tr><tr><td>2</td><td>Inactive</td></tr><tr><td>4</td><td>Expired</td></tr><tr><td>9</td><td>Suspended</td></tr></table>	Value	Description	1	Active	2	Inactive	4	Expired	9	Suspended				
Value	Description																			
1	Active																			
2	Inactive																			
4	Expired																			
9	Suspended																			
326	SecurityTradingStatus	Y	1	35	unsigned int	<table><tr><th colspan="2">Instrument trading state</th></tr><tr><th>Value</th><th>Description</th></tr><tr><td>2</td><td>Trading Halt</td></tr><tr><td>200</td><td>Closed</td></tr><tr><td>201</td><td>Restricted</td></tr><tr><td>202</td><td>Book</td></tr><tr><td>203</td><td>Continuous</td></tr></table>	Instrument trading state		Value	Description	2	Trading Halt	200	Closed	201	Restricted	202	Book	203	Continuous
Instrument trading state																				
Value	Description																			
2	Trading Halt																			
200	Closed																			
201	Restricted																			
202	Book																			
203	Continuous																			
2705	MarketCondition	Y	1	36	unsigned int	<table><tr><th colspan="2">Indicator for stressed market conditions.</th></tr><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Normal</td></tr><tr><td></td><td></td></tr></table>	Indicator for stressed market conditions.		Value	Description	0	Normal								
Indicator for stressed market conditions.																				
Value	Description																			
0	Normal																			
2447	FastMarketIndicator	Y	1	37	unsigned int	Not used														
1174	SecurityTradingEvent	N	1	38	unsigned int	Not used														

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description	
25155	SoldOutIndicator	N	1	39	unsigned int	Not used	
1227	ProductComplex	Y	1	40	unsigned int	Value	Description
						1	Simple Instrument
						5	Futures Spread
268	NoMDEntries	Y	1	41	Counter	Number of entries in Market Data message for MDEntryGrp.	
25021	Pad6	U	6	42	Fixed String	not used	
<MDInstrumentEntryGrp>						Variable size array, Record counter: NoMDEntries	
270	>MDEntryPx	N	8	48	PriceType	Price.	
271	>MDEntrySize	N	8	56	QuantityType	Quantity.	
269	>MDEntryType	Y	2	64	unsigned int	Type of market data entry.	
						Value	Description
						2	Trade
						4	Opening Price
						5	Closing Price
						7	High Price
						8	Low Price
						66	Trade Volume
						101	Previous Closing Price
						218	Open Interest
						334	Life Time Low
						335	Life Time High
277	>TradeCondition	N	2	66	unsigned int	May be set together with MDEntryType 2=Trade or 66=Trade Volume	
						Value	Description
						155	Midpoint price (BB)
						624	Trade At Close (TC)
25017	>Pad4	U	4	68	Fixed String	not used	
30000	>OILastUpdateTime	N	8	72	UTCTimestamp	This is filled when MDEntryType – 218 Open Interest. Indicates Last updated Open Interest Time	

Open Interest – Open Interest Indicates the Latest Open Interest value due to trade and Post Trade event processing

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U23	3	Fixed String	U23 = Market Data Instrument

28842	MarketDataType	10	1	unsigned int	10 = Single Instrument Snapshot See also MCX T7 EOBI Schema (XSD) file.
22	SecurityIDSource	M	1	Fixed String	M = Marketplace Marketplace assigned identifier.

4.18 Snapshot Order = 13602

Each individual order or quote is represented as a Snapshot Order in a snapshot cycle on the MCX T7 Enhanced Order Book Interface snapshot channel. The format of the snapshot order enables participants to build the order book according to price-time priority.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description						
<MessageHeader>												
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message including this field.						
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX T7 EOBI message layout. Value: 13602 (MarketDataOrder, MsgType = U21)						
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.						
<Message Body>												
<OrderDetails>												
21008	Reserve2	Y	8	8	UTCTimestamp	Not used.						
1138	DisplayQty	Y	8	16	QuantityType	Quantity.						
54	Side	Y	1	24	unsigned int	Side of the order. <table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Buy</td></tr><tr><td>2</td><td>Sell</td></tr></table>	Value	Description	1	Buy	2	Sell
Value	Description											
1	Buy											
2	Sell											
40	OrdType	N	1	25	unsigned int	Used for cash market instruments on-ly. 1 = Market Order Used for cash market instruments only. <table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Market Order</td></tr></table>	Value	Description	1	Market Order		
Value	Description											
1	Market Order											
25021	Pad6	U	6	26	Fixed String	not used						
44	Price	N	8	32	PriceType	Price.						

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U21	3	Fixed String	U21 = Market Data Order
28842	MarketDataType	11	1	unsigned int	11 = Order Book Snapshot See also MCX T7 EOBI Schema (XSD) file.
279	MDUpdateAction	5	1	unsigned int	5 = Overlay

4.19 Instrument Info = 13603

Instrument Info – An Instrument Info message will be published for an instrument on the MCX T7 Enhanced Order Book Interface incremental and snapshot channel whenever there is a change in the daily price range of the instrument.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX EOBI message layout. Value: 13203 (MarketDataTrade, MsgType = U22)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.
5	ClosePrice	N	8	16	PriceType	Close Price
140	PrevClosePrice	N	8	24	PriceType	Previous Close Price
332	UpperDailyPriceLimit	N	8	32	PriceType	Upper price Limit
333	LowerDailyPriceLimit	N	8	40	PriceType	Lower Price Limit

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U22	4	Fixed String	Defines message type ALWAYS FIRST FIELD IN MESSAGE. (Always unencrypted) Note: A 'U' as the first character in the MsgType field (i.e. U, U2, etc) indicates that the message format is privately defined between the sender and receiver.
28842	MarketDataType	14	1	unsigned int	Type of public market data, e.g., Order Book Maintenance (=1), Order Book Execution (=2), Market Segment Snapshot (=9) etc. Valid values are available in the MCX EOBI Schema (XSD) file.

4.20 Trade Report = 13201

In case of Market Reset Events due to Technical Interruptions, Trade Report message is published for traded contracts.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a T7 EOBI message layout. Value: 13201 (Market-DataTrade, MsgType = U22)
34	MsgSeqNum	Y	4	4	unsigned int	Message sequence number, incremented per product across all message types.
<Message Body>						
<OrderDetails>						
48	SecurityID	Y	8	8	Signed int	Unique instrument identifier.
60	TransactTime	Y	8	16	UTCTimestamp	Transaction Timestamp.
32	LastQty	N	8	24	QuantityType	NA
31	LastPx	Y	8	32	PriceType	Price of this fill.

880	TrdMatchID	N	4	40	Unsigned int	NA
44	Price	N	8	44	PriceType	Price.
574	MatchType	N	1	52	unsigned int	NA
28610	MatchSubType	N	1	53	unsigned int	NA
2667	AlgorithmicTradeIndicator	N	1	54	unsigned int	NA
25016	Pad1	U	1	55	Fixed String	Not used.
277	TradeCondition	N	2	56	Unsigned int	NA
25021	Pad6	U	6	58	Fixed String.	Not used.

Implied Message Constants

These constant values are to be considered as part of the above message, although they are not transmitted.

Tag	Field Name	Field Value	Length	Data Type	Description
35	MsgType	U22	3	Fixed String	U22 = Market Data Trade
28842	MarketDataType	10	1	unsigned int	Trade Report
22	SecurityIDSource	M	1	Fixed String	M = Marketplace assigned identifier.
279	MDUpdateAction	1	1	unsigned int	New

5 Index

5.1 Index Info = 13604

Index Info – This message is published to communicate computed price of Unique Instrument identifier for index. This message does not contain Packet header and sent on Index stream.

Tag	Field Name	Req'd	Len	Ofs	Data Type	Description
<MessageHeader>						
9	BodyLen	Y	2	0	unsigned int	Number of bytes for the message, including this field.
28500	TemplateID	Y	2	2	unsigned int	Unique identifier for a MCX EOB message layout. Value: 13604 (MarketDataTrade, MsgType = U22)
34	MsgSeqNum	Y	4	4	unsigned int	unique non contiguous sequence number assigned for each packet which is always greater than previous message sequence number
<Message Body>						
48	SecurityID	Y	8	8	signed int	Unique instrument identifier.
270	MDEntryPx	Y	8	16	PriceType	Computed Index Value
30000	LastUpdateTime	N	8	24	UTCTimestamp	Indicates last updated Index value Time (in nano-seconds)